

# Kuan-Cheng (James) Ku

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## Education

- National Taiwan University** *Master in Information Management* Feb 2026 – Present  
**National Taiwan University** *Bachelor in Information Management (3.88 GPA)* Sep 2021 – June 2025
- Key courses include Data Structures, Algorithms, Database, Web Programming, Deep Learning, Information Retrieval & Text Mining, Cloud Native Application, Systems Analysis & Design, Management of Hi-tech Company
  - Teaching assistant for 1000+ students across 7 courses, including Web Programming (EE), Web APP Development (IM), Machine Learning, Data Structures, Systems Analysis & Design, Calculus, and Marketing Management (EMI course).
  - Research Assistant @ Prof. Hsin-Min Lu's BAEIR Lab, focusing on LLMs applications in job-resume matching.
  - Research Assistant @ NTU Hospital Peritoneal Dialysis Smart Healthcare Project.

## Experience

- Research Development Intern, Microsoft Taiwan** *TSMC OnePlatform Team, ISD* Apr 2025 – Jan 2026
- Led e2e testing from design to acceptance, executing 400+ test cases and coordinating QA-client validation.
  - Identified 150+ bugs, built frontend analysis POC, and led cross-vendor meetings to accelerate bug resolution.
  - Authored 300+ page LaTeX manual and delivered critical training sessions instrumental to successful project closure.
- Contract Developer, National Chung Hsing University** *NCHU Buddy Program* Apr 2025 – Sep 2025
- Architected and developed a full-stack student management dashboard using Next.js, Supabase, and Google Cloud API.
  - Built task management, dynamic forms, real-time analytics, notifications, and Google Drive integration with RBAC.
  - Shipped production platform serving 300+ MAU, completed client acceptance, and successfully concluded project handoff.

## Achievements

- Taiwan Digital Innovation Awards 2025: Honorable Mention** [🔗 RenderResume](#) [📺 Demo](#) Jun 2025 - Oct 2025
- Built RenderResume, a resume platform featuring AI analysis and chat agent with in-context editing capabilities.
  - Gained 300+ users & 3,000+ uses during Beta, validating product-market fit as strategic entry into job matching market.
  - Selected as Top 7 finalist from 200+ teams nationally.
- TSMC IT CareerHack 2024: 2nd Place** [🔗 GitHub](#) [📜 Certificate](#) Feb 2025
- Developed an AI maintenance toolkit with web and VSCode interfaces using Vertex AI, FastAPI and LangChain
  - Rapidly prototyped and deployed the full-stack solution with GitHub integration on GCE and GKE within 2 days.

## Projects

- One-Click Outfit** [🔗 GitHub](#) [📺 Demo](#) [📄 Report](#) [📖 Poster](#) [📄 Slides](#) Jan 2024 – Jan 2025
- Led team of 4 to build AI fashion recommendation platform using Next.js and Flask for personalized styling.
  - Improved performance 2.4x by migrating to PostgreSQL with vector indexing and Docker deployment on VPS.
- SARS: Social Advertising Recommendation System** [📄 Report](#) [📄 Slides](#) Nov 2023 – Jan 2024
- Developed ad engine integrating BERT-based comment scoring and Dijkstra-based user subgraph modeling.
  - Optimized retrieval parameters using Gurobi solver and systematic backtesting.
- Customized IT Solution for ShiJiou Cafe** [📄 Report](#) [📄 Slides](#) Mar 2023 – May 2023
- Developed automated customer flow system for local coffee shop, addressing wait times and queue inefficiencies.
  - Built online menu, LINE chatbot queueing, and unified backend using React, Express, Firebase & LINE Messaging API.
- StreamBench Agent: Continuous Improvement of LLM** [📺 Presentation](#) [📄 Report](#) Oct 2024 – Dec 2024
- Built StreamBench-inspired pipeline for medical diagnosis LLM, improving accuracy from 50.17% to 73.64%.
  - Engineered prompting techniques, including clinical roleplay, structured reasoning (CoT) and negative example learning.

## Volunteering and Leadership

- Lecturer, Taipei Municipal Datong High School** *Information Research Club* Sep 2022 – Jan 2026  
**Lecturer, Taipei Municipal Zhongshan Girls High School** *Information Research Club* Feb 2024 – Jan 2026

- Lecturer, NTU CSIE Sprout Program**
[▶ Example Lecture](#)
[📄 Example Slides](#)
Dec 2021 – Jun 2024
- Designed and delivered C++ programming courses to foster pre-univ students’ interest in computer science, build foundations for APCS preparation, and guide career exploration in CS fields.
- Head of IT, NTU Information Management Student Association**
Jun 2022 – Jan 2024
- Led development of a faculty interview platform to support students’ bachelor project advisor selection.
  - Co-administered Linux workstation systems and Google Workspace for the department.
- Co-founder, NTUIM Competitive Programming Team**
[@ Instagram](#)
Jan 2022 – Jun 2023
- Co-founded the team, recruited 2 lecturers and 36 members, and raised NT\$100K worth of rental space sponsorship.
  - Co-developed materials and delivered sessions on data structures and algorithms for competitive programming.
  - Regularly organized in-person, team-based ICPC-format contests on Codeforces for hands-on practice.
- Chief Organizer, 2023 NTUIM Programming Contest: PDAO**
[📘 Facebook](#)
Dec 2022 – May 2023
- Led organization of a nation-wide ICPC-format programming competition with cash flow exceeding NT\$100K.
  - Innovatively opened and coordinated synchronization of two off- and one on-campus venues to provide better experience.
  - Successfully recruited 81 teams comprising 226 participants, including 10+ professional competitive programming teams.
  - Achieved a post-competition satisfaction rating of 4.21/5, demonstrating strong participant engagement.
- Deputy Organizer, 2024 NTUIM Camp**
[@ Instagram](#)
[▶ Youtube](#)
[📖 Brochure](#)
Jun 2023 – Jan 2024
- Revived a 5-day camp from 4-year pandemic suspension by recruiting 60+ staff with cash flow exceeding NT\$800K.
  - Oversaw marketing and design, a cross-functional department of 10, comprising art, publicity and photography teams.
  - Led merchandise development, multi-channel event promotion, brand establishment, and comprehensive documentation, gaining 1K+ followers and 500K+ impressions on social media within 6 months, contributing to 200+ registrations.
- Dev Lead & Event Planner, NTU Art Festival 2024 Riddle City Team**
[🔗 Legacy](#)
Jan 2024 – Jun 2024
- Led dev team of 3 to build treasure hunt game integrating web and physical campus riddles using Next.js and Supabase.
  - Bridged planning, design, and dev teams to launch game engaging 200+ users with 600+ gameplay hours.
- Member, NTU Photography Club Exhibitions Department**
[🔗 Online Exhibition](#)
Jul 2023 – Dec 2023
- Organized promotion booth and biannual exhibitions providing members with platforms to showcase their works.
  - Developed an online exhibition website using React.js and Firebase to expand exhibition reach and accessibility.

Skills and Languages

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|---------------------------------------------------------------------------------------|------------------------|
| <b>Languages:</b> Python, C/C++, HTML5, CSS3, JavaScript, TypeScript, SQL             | <b>Chinese:</b> Native |
| <b>Frameworks:</b> Express, React, Next, Tailwind, LangChain, Flask, FastAPI, Pytorch | <b>English:</b> Fluent |
| <b>Database:</b> PostgreSQL, Supabase, Cloudflare D1 & R2, MongoDB, Firebase          | TOEIC 950              |
| <b>Tools:</b> Git, Docker, GCP, Vercel, Cloudflare, Azure DevOps, Playwright          |                        |